

Algorithmic Trading Strategy Optimization

Using Genetic Algorithms — Detailed Report

Metric	Training Set	Test Set (GA)	Buy & Hold (Test)
Total Return	+1377733.27%	-7.54%	-17.76%
Sharpe Ratio	1.8767	-4.0000	4.6350
Max Drawdown	99.32%	99.81%	99.09%
Number of Trades	5	2	N/A (hold)

Evolved Chromosome — Best Individual:

rsi_oversold

: 31.4133

rsi_overbought

: 78.1007

sma_ratio_buy

: 0.9635

sma_ratio_sell

: 1.0499

macd_diff_buy

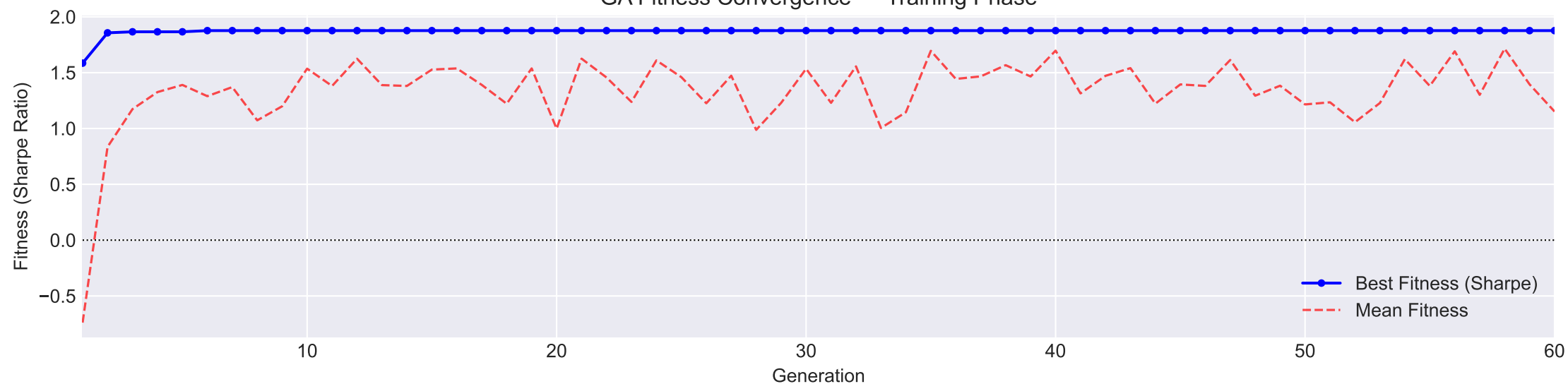
: -3.1572

macd_diff_sell

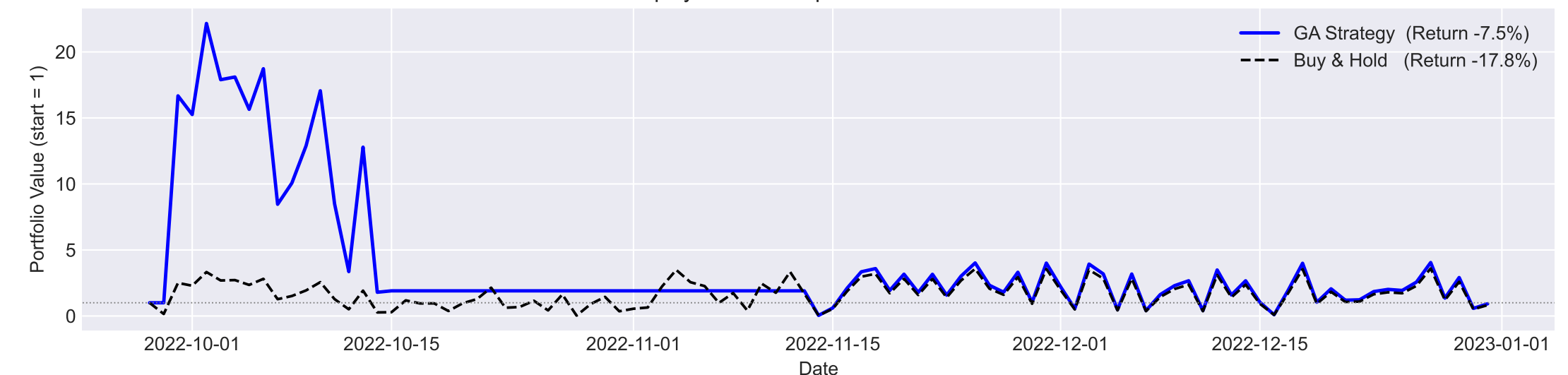
: -4.1366

Algorithmic Trading GA — Results Dashboard

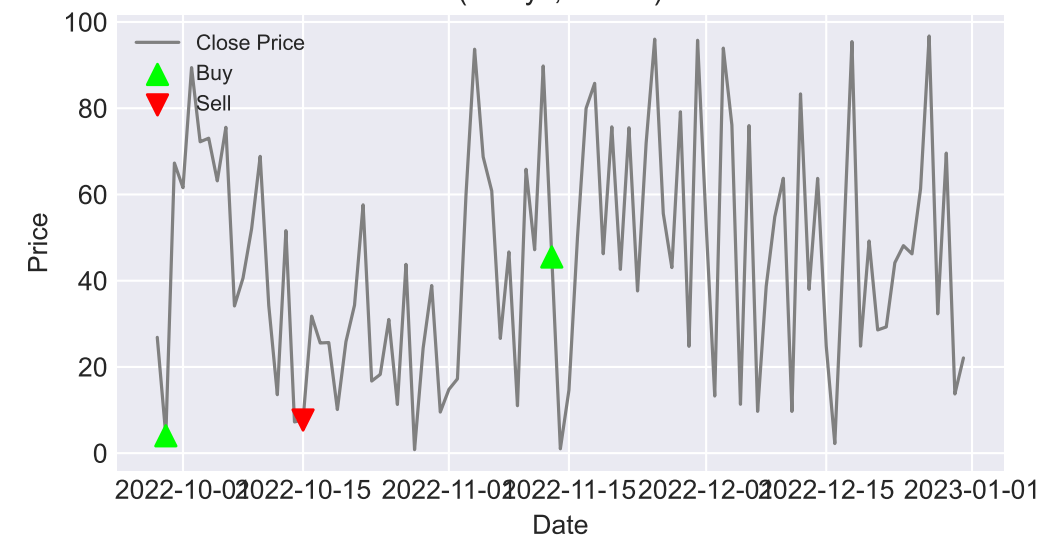
GA Fitness Convergence — Training Phase



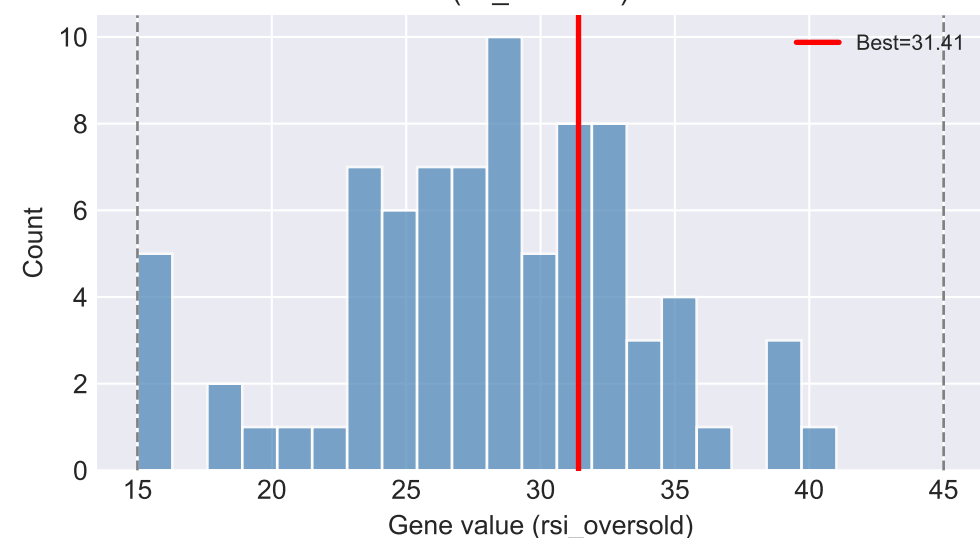
Equity Curve Comparison — Test Set



Buy/Sell Signals — Test Set (2 buys, 1 sells)

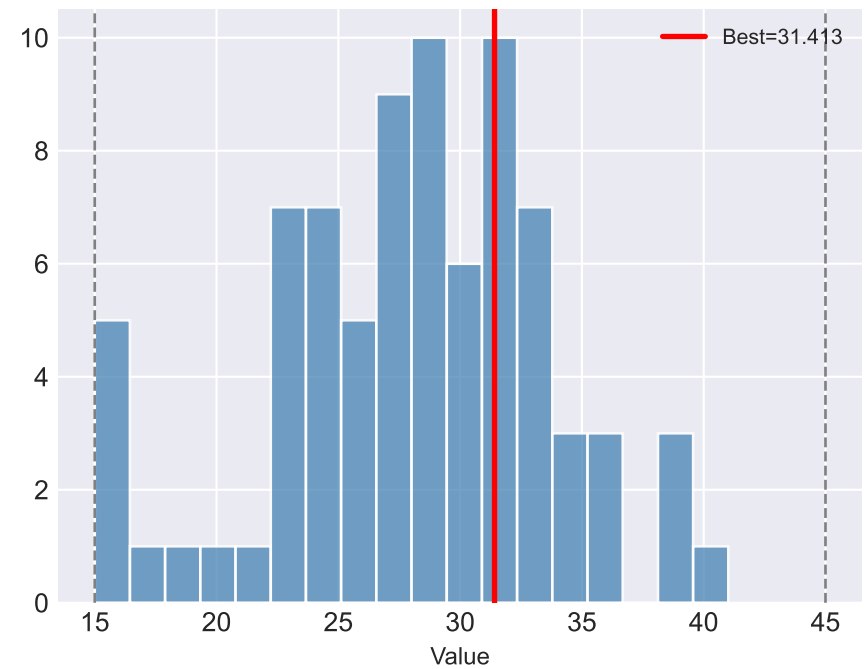


Final-Population Gene Distribution (rsi_oversold)

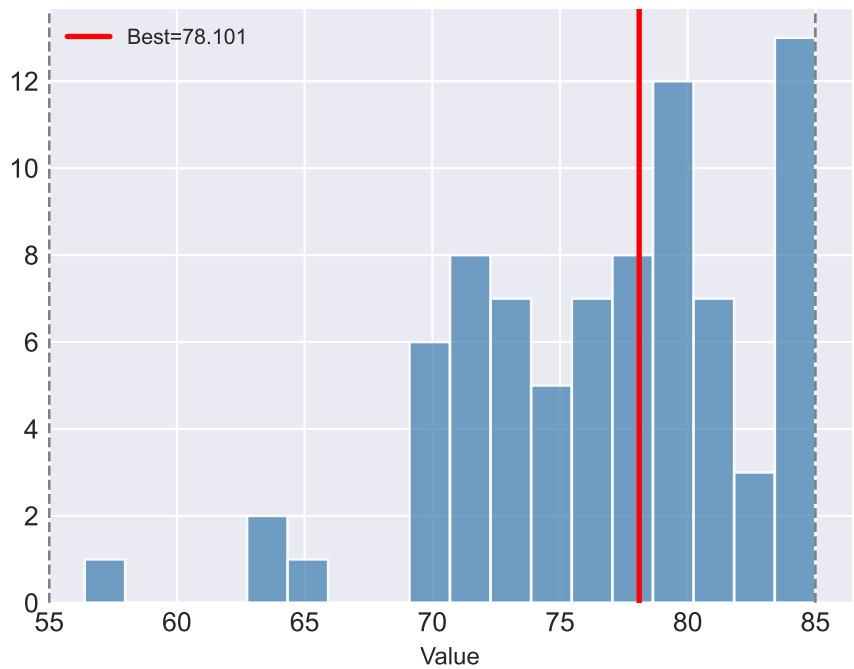


Distribution of Evolved Gene Values — Final Population

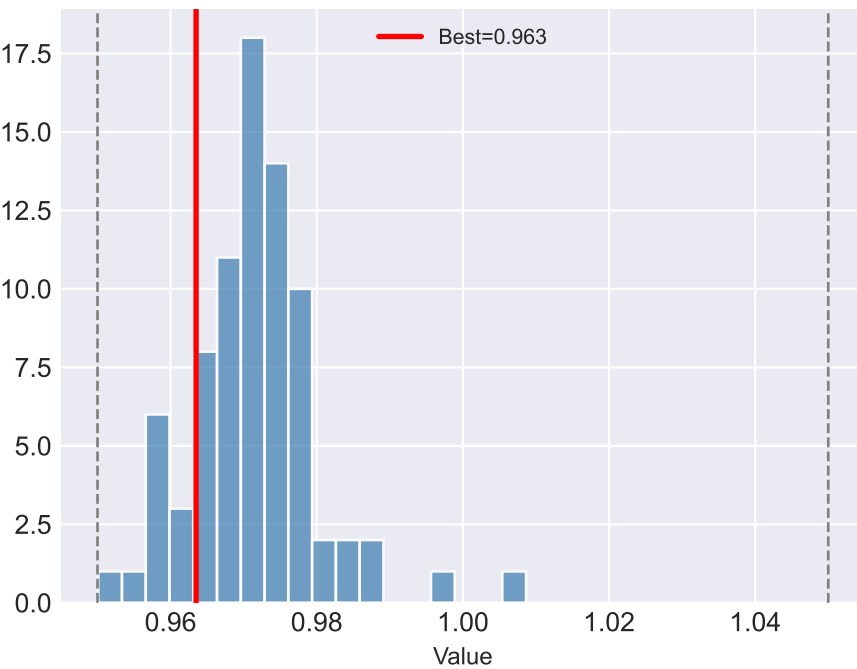
Rsi Oversold



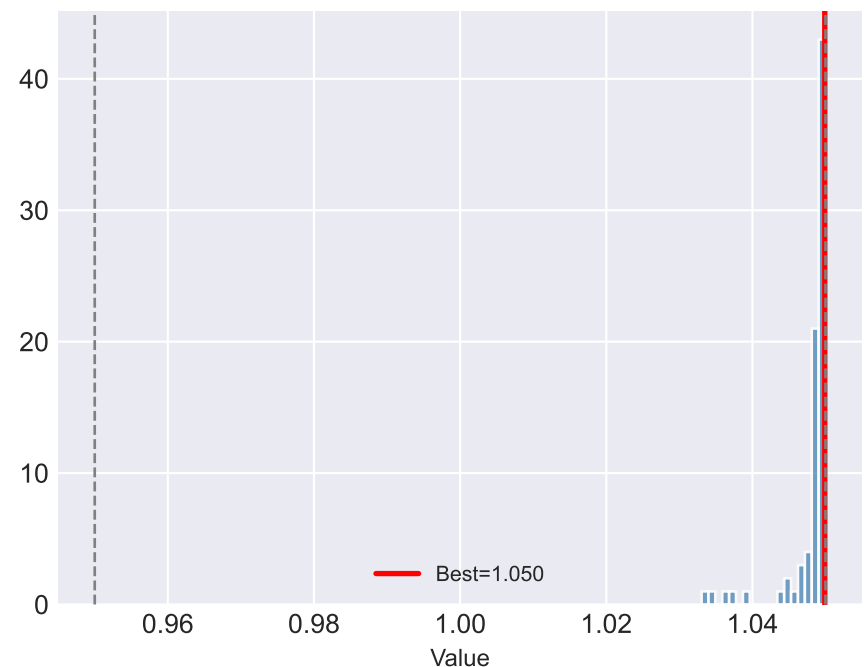
Rsi Overbought



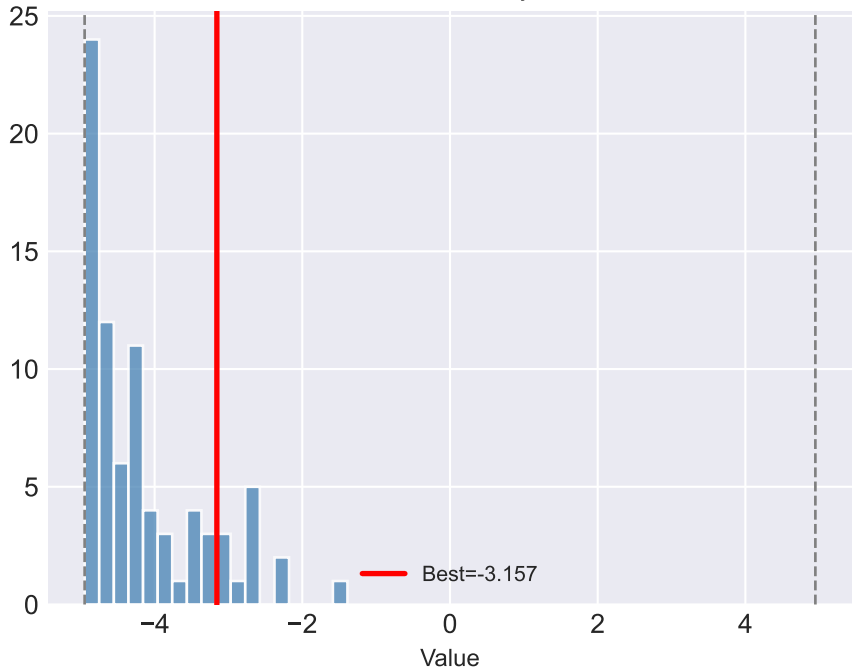
Sma Ratio Buy



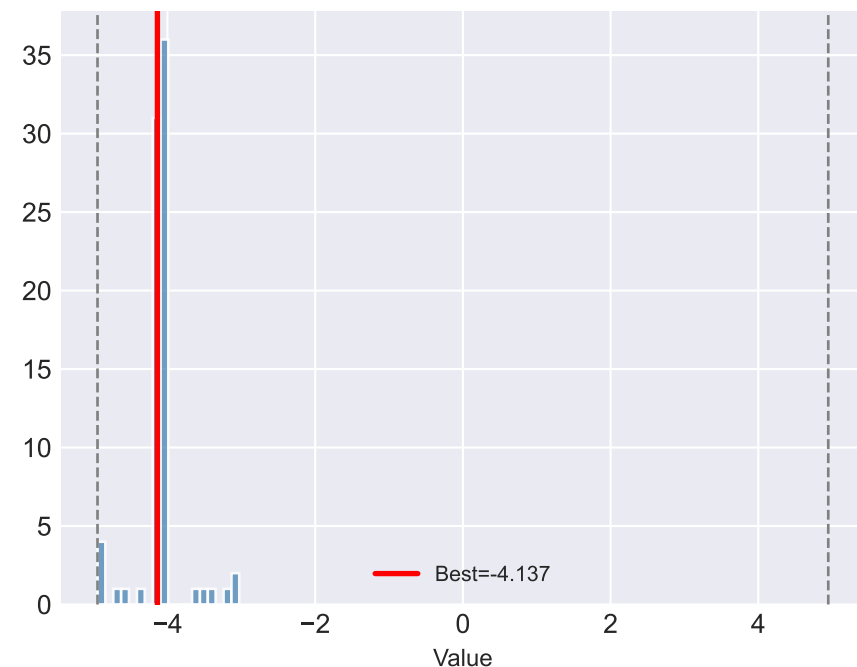
Sma Ratio Sell



Macd Diff Buy



Macd Diff Sell



Algorithm Details & Design Notes

GENETIC ALGORITHM DESIGN

CHROMOSOME – 6 real-valued genes (decision thresholds)

rsi_oversold	[15–45]	Buy when RSI falls below this
rsi_overbought	[55–85]	Sell when RSI rises above this
sma_ratio_buy	[0.95–1.05]	Buy when SMA20/SMA50 > ratio
sma_ratio_sell	[0.95–1.05]	Sell when SMA20/SMA50 < ratio
macd_diff_buy	[-2σ , $+2\sigma$]	Buy when MACD–Signal > threshold
macd_diff_sell	[-2σ , $+2\sigma$]	Sell when MACD–Signal < threshold

SIGNAL LOGIC – 2-of-3 MAJORITY VOTE

BUY if ≥ 2 of {RSI<oversold, SMA_ratio>buy, MACD_diff>buy_diff}

SELL if ≥ 2 of {RSI>overbought, SMA_ratio<sell, MACD_diff<sell_diff}

GA HYPERPARAMETERS

Population size	: 80		Generations	: 60
Elitism	: 10%		Tournament k	: 4
Crossover	: BLX- α ($\alpha=0.30$)			
Mutation	: Gaussian, rate=20%, $\sigma=10\%$ of gene range			

FITNESS FUNCTION

fitness = Sharpe(all daily returns) – max(0, MDD–0.30)×3

0 trades → –10 (hard penalty) | <3 trades → –5 + 0.5×trades

Uses ALL returns incl. flat days – penalises idle strategies.

DATA SPLIT

Training : 70% (221 bars, 2022-02-19 → 2022-09-27)

Test : 30% (95 bars, 2022-09-28 → 2022-12-31)

ORIGINAL BUGS FIXED

1. Population of identical function refs → 6-gene float chromosome
2. Crossover was a no-op → BLX- α real-valued blend
3. Mutation reassigned same fn → Gaussian gene perturbation
4. AND-logic generated zero trades → 2-of-3 majority vote
5. Fitness always 0 → Sharpe + drawdown penalty
6. No selection pressure → Tournament + elitism